# Agentic AI in mathematical research

# A brief disclaimer

I am not an expert in this field. I have been exploring it out of curiosity and to stay up to date.

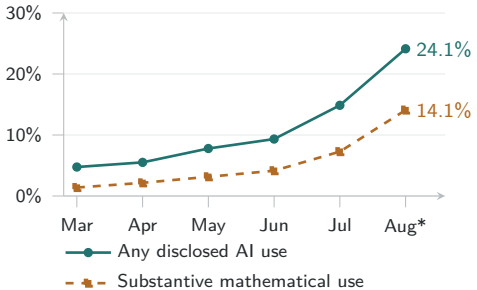
This talk neither argues for nor argues against using AI to write papers.

- A brief history and overview of mathematical research with LLMs, focusing on agentic approaches.
- What the literature tells us about successful approaches and their limitations.
- A case study where I build an AI research team to write a paper.

# AI use and mathematical output

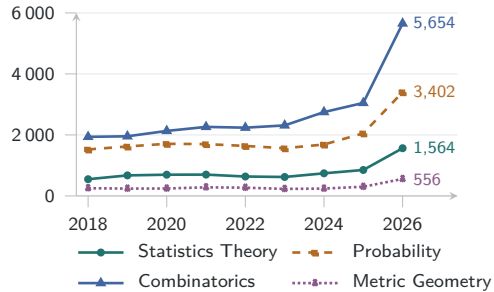
## AI disclosures on arXiv

Share of math-related submissions, 2026




## Submissions by field

New preprints, January–August each year

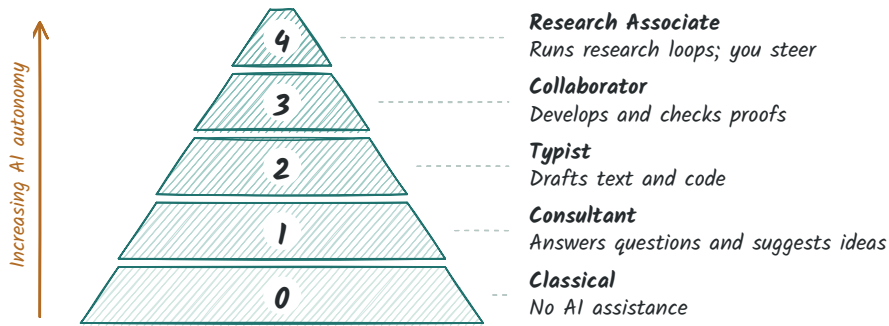



Left: 3,575 of 32,944 preprints disclosed AI use. *August covers 1–20.
Right: category counts for published arXiv papers.

Jin, Ke & Sui (2026), Table 1  Explore arXiv Math Pulse  GitHub source.

# (How) have you used AI in your research?

Which is the highest level you have used?




Adapted from Zimmer et al. (2026), *The Agentic Researcher*, Section 2.

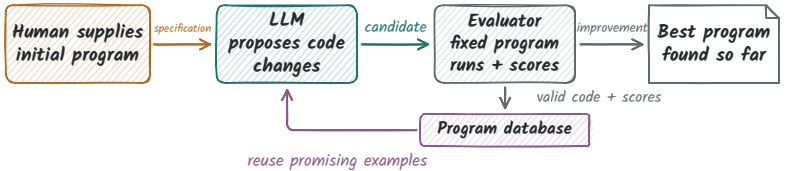
# A short history of math research with LLMs

## Agentic research systems over time

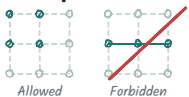
A brief history, with a focus on mathematics.

# **FunSearch** Romera-Paredes et al., *Nature* (2024)

An evolutionary approach with fixed evaluation criteria implemented in computer programs.




## Larger cap sets



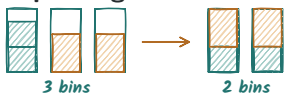

496 → 512

points in 8 dimensions

No three distinct points on a line in $\mathbb{F}_3^n$.

A new construction, not a proof of maximality.

## Online bin-packing heuristics



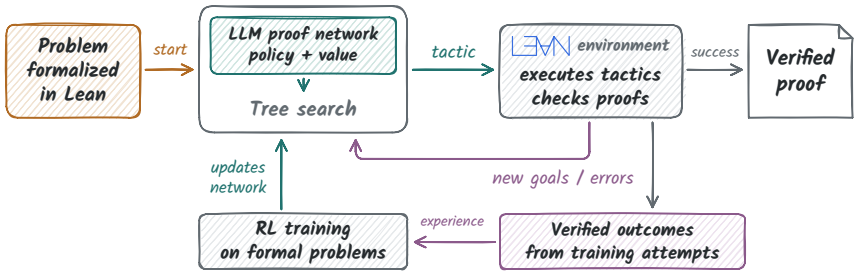

Use as few fixed-capacity bins as possible; place each item on arrival.

AlphaEvolve (2025) later extended this approach: $4 \times 4$ complex matrix multiplication with 48 scalar multiplications, versus Strassen's 49.

Workflow: Fig. 1; results: Fig. 4 and Table 1. Romera-Paredes et al. (2024).

# **AlphaProof** Hubert et al., *Nature* (2026)

Another way to verify and evaluate mathematics is through formal verification in Lean.




**Search:** the LLM proposes tactics; Lean returns new proof states.

**Learning:** The model learns from proof attempts. For harder problems, it practises on similar ones.

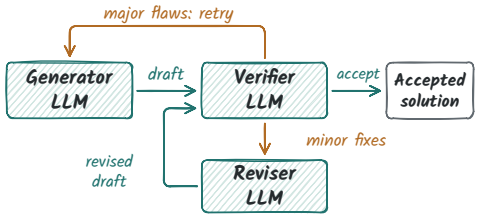
IMO 2024: silver-medal level — 28/42 points with AlphaGeometry 2.
AlphaProof solved 3 problems; AlphaGeometry 2 solved 1.

Adapted from Figs. 1–2 and the AlphaProof section. *Nature* 651, 607–613 (2026).

# **Aletheia** Feng et al., arXiv (2026)

We can also use an independent LLM for verification. This does not guarantee correctness.

### Gemini Deep Think agents




LLM judgment; no formal guarantee (Fig. 1).

### Reported mathematical results

- Erdős: 2 full + 2 partial new solutions from 700 problems, assessed by experts.

And other mathematical achievements.

### Other LLM research systems

**The AI Scientist** *Nature* (2026)
Ran ML experiments and wrote a paper that passed a workshop's first review round.

**Google Co-Scientist** *Nature* (2026)
Agents proposed and refined leukaemia drug combinations, validated in laboratory experiments.

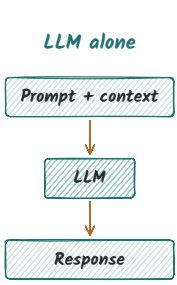
**Virtual Lab** *Nature* (2025)
LLM agents designed SARS-CoV-2 nanobodies with binding confirmed in laboratory tests.

**OpenAI: Navier–Stokes** report (2026)
Different approaches explored with 10,000 agents and human steering.

# Agentic systems

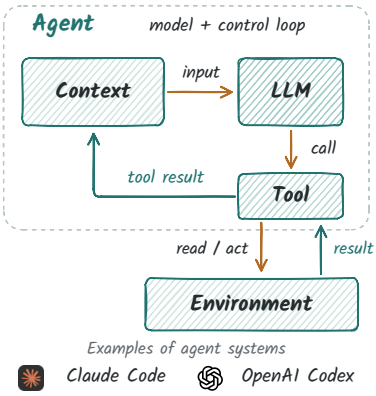
From LLMs to research teams

# Environment, context, tools and agents




$$Y = f_\theta(p, c; \xi)$$

$p$: prompt; $c$: context
$\theta$: fixed model
$\xi$: randomness




**Environment**
The workspace an agent can access: files, data, code and external services.

**Context**
The information given to the LLM now:
instructions, messages and tool results.

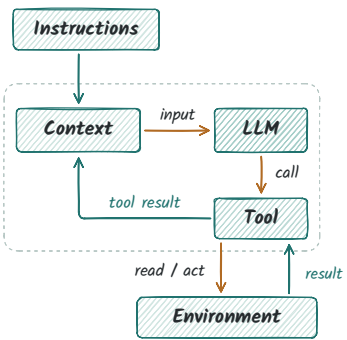
**Tool**
An interface to observe or change the environment, e.g. search or Python.

**Agent**
An LLM-based system that uses tools and chooses its next step from results.

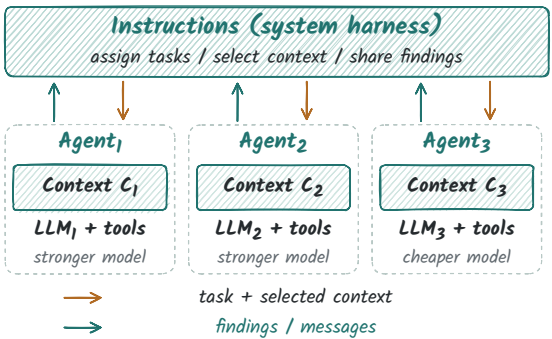
Based on OpenAI's Codex harness documentation and Claude Code's agentic loop.

# Instructions and multi-agent systems

## One agent




## Several agents




Each agent has its own context. Instructions specify its task and how it should work.

Same or different models. Sources: ReAct (Yao et al., 2022); AutoGen (Wu et al., 2023).

# Why manage context and communication?

An LLM produces text $Y = f_\theta(p, c; \xi)$ whose distribution depends strongly on the context.

**1. Keep context focused** Liu et al. (2024); Modarressi et al. (2025)
**Problems:** context distraction and context clash.
Give each agent the information it needs; long histories can obscure relevant details.

**2. Give review a fresh context** Sharma et al. (2023); Dhuliawala et al. (2023)
**Problems:** context poisoning and context confusion.
Review the draft and criteria without the writer's history; verify independently.

**3. Match the model to the task** Ong et al., *RouteLLM* (2024)
Routing between stronger and cheaper models can improve the balance of cost and quality.

**4. Share findings and avoid duplicate work** Du et al. (2023); Wang et al. (2025)
Share useful results and redirect work that adds no new information.

**5. Explore distinct approaches independently** Zhu et al. (2026)
Especially important for mathematics: different starting points can reveal new proof strategies.

Effects depend on the task and models. Fresh-context review is a design inference from bias and verification studies.

# Agentic research: a worked example

What do we need?

- An agent system: in my case, Codex.
- A precise mathematical problem statement.
- Relevant related work: definitions, known results and useful methods.
- `instruction.md`: the task, constraints and expected output.
- `skills.md`: an index of reusable methods and checks.
- `AGENTS.md`: standing project instructions for the agents.

Here, `instruction.md` and `skills.md` are our project documents.
Each actual Codex skill has its own `SKILL.md`.

# Point processes: a Poisson example

A **simple point process** $X$ is a random set of distinct points in $\mathbb{R}^d$, finite in bounded regions.

Its count in a region $B$ is $X(B) = \#(X \cap B)$.

**Stationarity**
Shifting all points leaves the distribution unchanged:

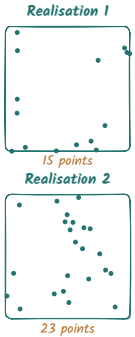
$$X + t \overset{d}{=} X \quad \text{for every } t \in \mathbb{R}^d.$$

**Homogeneous Poisson process, intensity** $\lambda > 0$

$$X(B) \sim \text{Poisson}(\lambda|B|), \qquad \mathbb{E}[X(B)] = \lambda|B|.$$

Here $|B|$ is the volume of $B$; $\lambda$ is the mean number of points per unit volume.

Counts in disjoint regions are independent.




Unit-area windows; $\lambda = 20$.
On average 20 points per window.

Baddeley, Sections 1.4–1.6. Pictures show independent simulated realisations in two dimensions.

# Product densities: counting pairs

For $n \geq 1$, the **factorial moment measure** is defined by

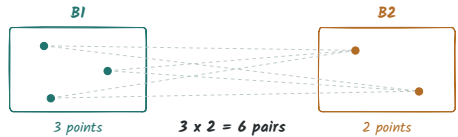
$$\alpha^{(n)}(A) := \mathbb{E} \sum\nolimits_{(x_1,\ldots,x_n)\in X^n_{\neq}} \mathbf{1}_A(x_1,\ldots,x_n), \quad A \subseteq (\mathbb{R}^d)^n \text{ Borel.}$$

Here $X^n_{\neq}$ consists of all ordered tuples of distinct points of $X$.

If $\alpha^{(n)}$ has a density, its **product density** $\rho^{(n)}$ satisfies

$$\alpha^{(n)}(A) = \int_A \rho^{(n)}(x_1,\ldots,x_n)\,\mathrm{d}x_1 \cdots \mathrm{d}x_n.$$

**Example: pairs in two disjoint regions.**
$X(B_1)X(B_2)$ counts pairs with one point in each region.
Its expectation is $\alpha^{(2)}(B_1 \times B_2)$.




For small regions around $x_1, x_2$, where the density varies little:

$$\mathbb{E}[X(B_1)X(B_2)] \;\approx\; \rho^{(2)}(x_1,x_2)\,|B_1|\,|B_2|.$$

Baddeley, Section 2.3; Last and Penrose, Section 4.4.

# Product densities: comparison with Poisson

For a homogeneous Poisson process, independence gives

$$\rho^{(2)}(x_1, x_2) = \lambda^2, \qquad \rho^{(n)}(x_1, \ldots, x_n) = \lambda^n.$$

For a stationary process with intensity $\lambda > 0$, compare its product density with this reference:

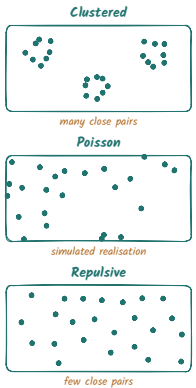
$$g_n(x_1, \ldots, x_n) = \frac{\rho^{(n)}(x_1, \ldots, x_n)}{\lambda^n}.$$

This is the *n*-**point correlation function**.

Compared with Poisson at the same intensity:

$g_n > 1$: more such tuples.
$g_n < 1$: fewer such tuples.
$g_n = 1$: the same local tuple density.

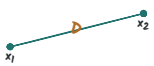
$n = 2$: pairs; $n = 3$: triangles; $n = 4$: tetrahedra.




Poisson simulated; other patterns schematic. Comparisons are local; $g_2 = 1$ alone does not imply Poisson.
Baddeley, Section 2.3; Last and Penrose, Section 4.4.

# Step 1: the precise problem

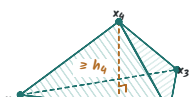
Fix $M, \lambda_0, h_3, h_4 > 0$.
Find one stationary, simple point process $X$ on $\mathbb{R}^3$ with intensity $\lambda_0$, finite local moments through order four and product densities.



Distance D



Area A



Volume V

For some constants $C_2, C_3, C_4 > 0$, require simultaneously:

$$g_2(x_1, x_2) = \frac{C_2}{D}, \qquad g_3(x_1, x_2, x_3) = \frac{C_3}{A}, \qquad g_4(x_1, x_2, x_3, x_4) = \frac{C_4}{V}.$$

Almost everywhere on nondegenerate configurations of diameter $\leq M$, with every triangle altitude $\geq h_3$ and every tetrahedron altitude $\geq h_4$.

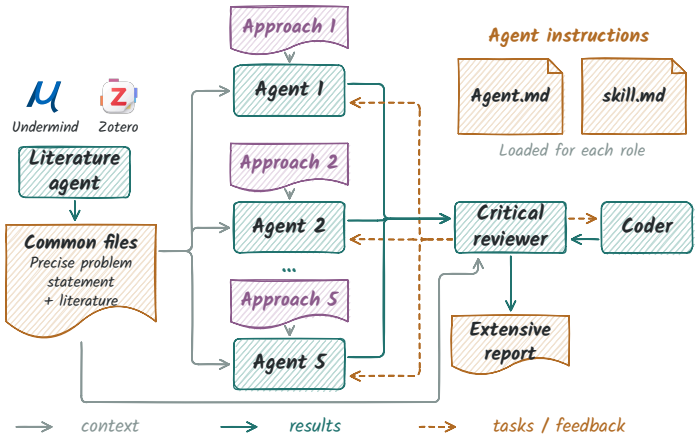
The conjecture is motivated by observed spatial correlations in earthquake catalogues.

Kagan and Knopoff (1980); Kagan (1981): three-point and four-point studies.

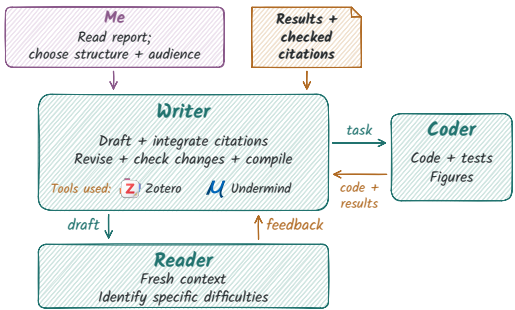
# Step 2: configuration/agent files and execution

**Research approaches**

1. Poisson cluster process
2. Cox process
3. Hawkes process
4. An approach of the agent's choice
5. Prove non-existence



# Step 3: Writing the paper




| | Calculations | Writing |
|---|---|---|
| Tokens | 132.5M | 5.78M |
| Time | 3 h 47 min | — |
| Price | $78.81 | $9.44 |

M = million tokens, including cached input.
API-equivalent prices in USD.
Writing time unavailable in the saved summary.

# Results and personal feedback

## Poisson cluster model

Homogeneous Poisson parents $\Pi$ each generate 0–3 children, with jointly chosen offsets $C_z$:

$$X = \bigcup_{z \in \Pi} \big( \{z\} \cup (z + C_z) \big).$$

$C_z$ contains only child offsets. The parent is retained, giving 1–4 points per cluster.

**How many points?** The solution gives a probability distribution over cluster sizes $j = 1, \ldots, 4$.

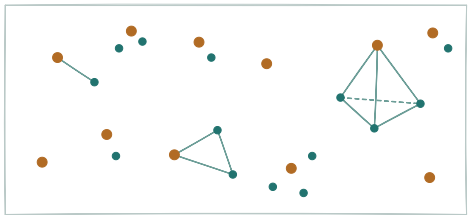
**Where are they placed?** For $j \geq 2$, the normalised density

$$\frac{F_j(z, x_1, \ldots, x_{j-1})}{Z_j}$$

describes how the $j - 1$ children are jointly distributed around the parent $z$.

## Personal feedback

| Aspect | Feedback |
|---|---|
| Mathematical correctness | Positive |
| Research and process documentation | Positive |
| Code writing | Positive |
| Text writing | Medium |
| Paper structuring | Negative |
| Mathematical explanation | Medium |



# Future of agentic research

## What I expect to change

Stronger models can increasingly plan, divide work into tasks, use tools and delegate to subagents.

For a precise mathematical problem, one initial prompt may increasingly suffice. This can still mean substantial autonomous work, with no guarantee of success.

As platforms provide more harness functions, we will have less to build ourselves.

I also expect explanations to become less sloppy and more useful for learning.

## Where a custom harness still helps

- Keep the exact target, hypotheses and criteria for success fixed.
- Require separate review and formal verification.
- Keep human checkpoints for direction and interpretation.
- Record results, failed approaches, sources and dependencies.
- Set compute budgets, stopping rules and model choice.
- When writing papers with AI, set your preferences for writing and illustration styles.

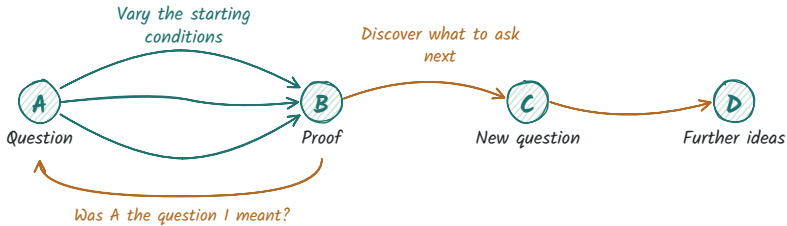
**A reported example.** For its October release, OpenAI says almost all results came from one prompt to one agent. Some may have needed retries; the full workflow has not been independently reproduced.

Scientific American, October 2026; OpenAI release details.

# Personal take

I find this exciting: an elegant solution, with reasoning that feels human.
Still, something is missing when I do not discover the ideas myself.

Faster proofs let me explore more. Research was never a straight line.




Reaching B only helps me if I understand the mathematics.

**My concern as a PhD student:** outsourcing too much may bypass the learning.
A PhD is still an education. Producing results and developing understanding can come apart.

Related: *A Severe Misalignment of AI in Mathematics* (joint declaration, 11 September 2026).

# Thank you!

Questions and discussion

Slides and, later, the experiments:

github.com/MathisRost/agentic-ai-math

